ADNAN SHAHZAD

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer specializing in the MERN stack, with a strong concentration in backend architecture, scalable API development, and performance optimization. Skilled in designing RESTful APIs, architecting maintainable Node.js applications, and optimizing MongoDB schemas and query performance, alongside building responsive, component-driven interfaces with React. Applies structured, system-design-driven problem-solving to translate complex requirements into clean, testable, production-grade solutions. Committed to long-term maintainability and architectural clarity, with a continuous focus on backend systems, distributed architectures, and performance engineering. Seeking opportunities in backend engineering, full-stack development, and scalable system design.

TECHNICAL SKILLS

Languages:	JavaScript, TypeScript, PHP, Java, Python, SQL
Backend:	Node.js, Express.js, REST APIs
Frontend:	React.js, Redux, React Native, HTML5, CSS3
Databases:	MongoDB, MySQL
Architecture:	OOP, MVC, Design Patterns
Testing:	Unit Testing, Performance Testing
DevOps & Tools:	Git, GitHub, CI/CD, GitHub Actions
Cloud & Auth:	AWS, Firebase, JWT

PROJECTS

Hand-Drawn Shape Recognition System | *Python, NumPy, OpenCV, MediaPipe* **|** Demo Video

- Built a custom Convolutional Neural Network (CNN) entirely from scratch in NumPy, engineering the underlying math for convolutions, pooling, and backpropagation.
- Achieved ~76% accuracy across 8 geometric classes on a 20,000-image dataset, outperforming baseline KNN and Logistic Regression models.
- Integrated the trained weights into a real-time OpenCV desktop application that tracks a user's finger using MediaPipe to classify shapes drawn in the air on the fly.

Semester Survival (Website Base Game) | *Node.js, React, MongoDB* **|** Game live

- Created an interactive web-based RPG using React and Redux where players manage resources (Energy, Stress, GPA) and engage in tactical mini-games to survive 8 progressive difficulty levels.
- Built a custom game engine and battle system using Javascript and React state, featuring real-time feedback logs and dynamic UI updates styled with Bootstrap 5 and animated with Framer Motion.
- Developed a robust Node.js/Express backend backed by MongoDB to securely store user accounts via JWT and maintain persistent game states and global leaderboards.

Good Muslim - Interactive Learning Platform | *HTML5, CSS3, JavaScript, Firebase, REST API* **|** Live Website

- Developed a responsive web application providing an immersive Quran reading and listening experience using Vanilla JS and CSS.
- Integrated the Al Quran Cloud REST API to dynamically fetch Surahs, translations, and high-quality audio recitations.
- Implemented secure user authentication and personalized ayah bookmarking functionality utilizing Firebase Auth and Firestore.
- Built a real-time Admin Dashboard using Firebase to monitor user growth, track daily engagement metrics, and manage user accounts.

Pass the Card (Card game) | *React Native, Expo, Firebase, Reanimated* **|** Game live

- Developed a cross-platform mobile card game using React Native and Expo Router, featuring secure user authentication and player statistics tracking.
- Engineered a real-time multiplayer state management architecture leveraging Firebase Realtime Database to ensure seamless, synchronized gameplay.
- Designed a premium, immersion-focused UI with dynamic "suit-aware" themes, glassmorphism elements, and liquid-smooth 60fps animations using React Native Reanimated.

GIT REPOSITORY

All personal and academic projects are available on GitHub: https://github.com/CoderHere07

EDUCATION

BS Software Engineering | GIFT Gujranwala Institute of Future Technologies *Nov 2023 – Expected Aug 2027*
Expected CGPA: 3.8

Relevant Courses: Data Structures, Algorithms, Object Oriented Programming, Databases, Advanced Programming, Discrete Structures, Linear Algebra, Probability and Statistics, Calculus, Android App Development, Game Development, Human Computer Interaction.

Diploma of Associate Civil Engineering | CTTC Christian Technical Training Centre *2020 –2023 Grade: "A"*